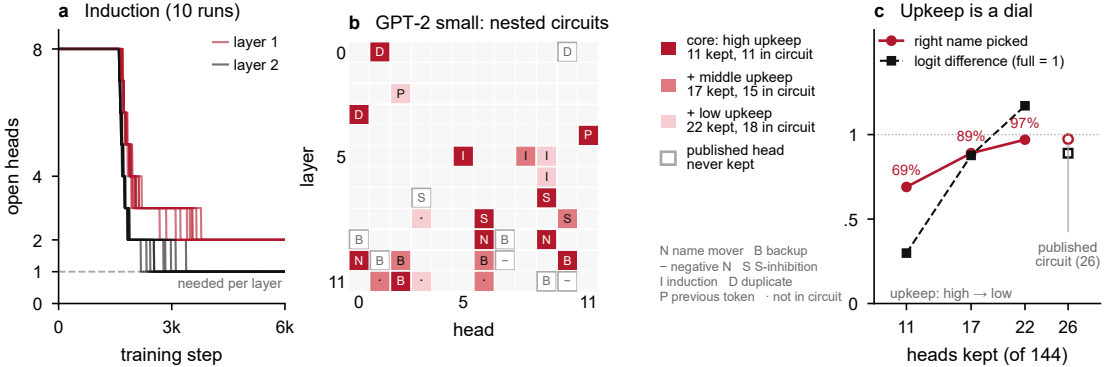
a
Induction (10 runs)
layer 1
layer 2
open heads
8
4
2
1
0
needed per layer
0
3k
6k
training step
b
GPT-2 small: nested circuits
layer
0
5
11
head
core: high upkeep
11 kept, 11 in circuit
+ middle upkeep
17 kept, 15 in circuit
+ low upkeep
22 kept, 18 in circuit
published head
never kept
N name mover B backup
– negative N S S-inhibition
I induction D duplicate
P previous token · not in circuit
c
Upkeep is a dial
right name picked
logit difference (full = 1)
69%
89%
97%
1
.5
0
published
circuit (26)
upkeep: high → low
11
17
22
26
heads kept (of 144)